YouTube Research Intelligence + Decision Engine

Production Agentic AI Architecture — Specification & Decision Document (LOCKED)

Status: LOCKED & FINALIZED	**Target MVP:** Phase 1 & 2 (Research + Decision)	**Date:** September 2026

> ***Core Architectural Principle:*** *Deterministic systems collect and calculate facts. AI agents interpret, reason, and generate. Humans approve consequential decisions. Autonomy is deliberately bounded, evidence is traceable, and failures are gracefully recoverable.*

1. Final 4-Tier System Architecture

Tier 4	**HUMAN LAYER**	Approval Gate 1 (Research & Topic validation), Override authority, Final strategic decisions.
Tier 3	**INTELLIGENCE LAYER**	Research Interpreter, Opportunity Reasoner, Content Strategist, Evidence Synthesizer. *(LLMs interpret evidence — they never invent telemetry).*
Tier 2	**DECISION LAYER**	Feature Engine, Outlier Engine, Topic Clustering, Competition Analysis, Question Gap Engine, Opportunity Engine, Confidence Engine, Evidence Ledger.
Tier 1	**DATA LAYER**	Channel/Video Scrapers, Transcript Collector, Comment Collector, Trend Sources, Data Normalization, Snapshot Storage with Temporal Provenance.
Core	**CENTRAL ORCHESTRATOR**	Controls workflow state, retries, task dependencies, data validation, error handling, and human checkpoints.

2. Locked End-to-End System Flow

Execution Pipeline:
USER REQUEST → REQUEST VALIDATOR → RESEARCH PLAN → DATA COLLECTION (Videos, Transcripts, Comments) → DATA NORMALIZATION → SNAPSHOT STORAGE → FEATURE EXTRACTION → AGE-COHORT OUTLIER ANALYSIS → TRANSCRIPT INTELLIGENCE → TOPIC CLUSTERING → COMPETITOR SATURATION ANALYSIS → QUESTION GAP GRAPH → OPPORTUNITY ENGINE (Observable Features) → EVIDENCE LEDGER → AI RESEARCH INTERPRETER → QUALITY & CONFIDENCE GATE → 5–10 GROUNDED OPPORTUNITIES → HUMAN APPROVAL GATE 1

3. Deterministic Performance & Outlier Engine

Mathematical Normalization: Raw views/hour is insufficient because it skews brand-new uploads. The system bins videos into **Age Cohorts** (0–6h, 6–24h, 1–3d, 3–7d, 7–30d, 30–90d, 90d+) and computes cohort median views and decay baselines.

Calculated Deterministic Telemetry:
• *views_per_hour & views_per_day* | • *views_vs_channel_median* | • *views_vs_age_cohort*
• *views_per_subscriber* | • *engagement_rate*, *like_rate*, *comment_rate*
• *age_adjusted_outlier* = Actual Views / Expected Cohort Views

Outlier Tiers (Deterministic): NORMAL (< 1.5x) | EMERGING (1.5x–3.0x) | STRONG OUTLIER (3.0x–5.0x) | EXTREME OUTLIER (> 5.0x). Each classification retains sample size and confidence.

4. Transcript Intelligence & Question Opportunity Graph

Structured Transcript Intelligence	Question Opportunity Graph
Transcripts are decomposed into narrative anatomy: • **Hook** (0–15s type: Question, Paradox, Bold Claim) • **Promise & Context** (15–60s expectation setting) • **Pattern Interrupts** (retention resets & pacing) • **Open Loops** (unresolved audience questions) • **Payoff & CTA** (conclusion delivery & cliff check) Aggregates recurring hooks and claims across top outliers.	Mines thousands of audience comments across competitors: 1. Comment Ingestion & Question Extraction 2. Semantic Normalization & Topic Clustering 3. Frequency & Volume Aggregation 4. Answered vs. Unanswered Rate in Competitor Videos 5. Discovery of high-demand, unaddressed audience gaps.

5. Observable Opportunity Formula & Evidence Ledger

No Black-Box Numbers: The Opportunity Score is calculated from observable features, then interpreted by the LLM:
`Opportunity = (Demand × Audience Fit × Outlier Evidence × Novelty) / (Competition × Production Cost)`

Evidence Ledger: Every opportunity links to concrete proofs: *Opportunity → Claim → Metric → Source Channel/Video → Snapshot Timestamp*. If data is insufficient, the system explicitly outputs **INSUFFICIENT EVIDENCE** rather than hallucinating.

6. Relational Database Schema with Temporal Provenance

Core Database Entities (PostgreSQL / SQLite):
• **Telemetry & Snapshots:** *channels, videos, video_snapshots, transcripts, transcript_segments, comments* (all tracking `scraped_at, observed_at, data_age`).
• **Intelligence & Analytics:** *topics, topic_clusters, keywords, competitors, features, outlier_scores, age_cohorts*.
• **Opportunities & Decisions:** *content_opportunities, opportunity_evidence, questions, question_clusters, evidence_ledger*.
• **Operations & Audits:** *research_runs, research_artifacts, agent_runs, human_reviews, audit_results, system_events*.

7. Bounded Agent Set & Implementation Scope

Approved LLM Agents (4 Only)	Strictly Excluded from MVP
1. **Research Planner:** Decides collection depth & sources. 2. **Research Interpreter:** Interprets structured facts. 3. **Opportunity Reasoner:** Synthesizes content gaps. 4. **Evidence Synthesizer:** Drafts human research report. *All math, scraping, clustering, DB ops are deterministic services.*	■ Automatic publishing ■ Autonomous thumbnail rendering ■ Automatic prompt rewriting / self-switching ■ Autonomous commenting / community actions ■ Auto-deletion or editing of live content *Future expansion hooks reserved for Content Factory.*

8. Next Immediate Implementation Steps

1. **Refactor Analytics Engine:** Implement Age Cohorts, Expected Decay Curve, and multi-feature outlier calculations in `analytics/metrics.py`.
2. **Build Structured Transcript Parser:** Create `research/transcript_intelligence.py` with Hook/Loop/Payoff extractor.
3. **Implement Opportunity Engine & Topic Clustering:** Build deterministic feature vector scorer in `research/opportunity_finder.py`.
4. **Deploy Central Orchestrator:** Create `orchestration/orchestrator.py` to coordinate pipeline & Gate 1 Human review.
5. **Update Schema & Verification:** Add snapshot tables in `database/models.py` and build comprehensive automated tests.